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First paleomagnetic evidence for a post-Eocene clockwise rotation of the Western Taurides thrust belt east of the Isparta reentrant (Southwestern Turkey)

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ABSTRACT

A paleomagnetic study was carried out on 23 sites sampled in Paleocene, Eocene and Lower Miocene sedimentary formations of the Western Taurides belt east of the Isparta reentrant. The paleomagnetic data, based on twelve suitable sites, record a coherent post-Eocene ~40° clockwise rotation throughout the area studied. We interpret this regional rotation as a semi-rigid rotation of the whole Western Taurides belt above a major deep décollement level. Together with previously published data, these new results suggest that the Isparta reentrant would result from diachronous opposite rotations of the two branches: first a late Eocene–Oligocene clockwise rotation of the Akseki–Beysehir Taurides initiating the Tauride arc, and then anticlockwise rotation of the Lycian Taurides, probably during Middle Miocene times.

1. Introduction

The Alpine Mediterranean belt constitutes a wide zone of continental collision made up of a succession of arcs. Many of these arcs surround extensional basins whereas a few others appear as deformed margins wrapping crustal blocks. In the Eastern Mediterranean, both of these types of structures are well illustrated by the Hellenic belt around the Aegean Sea and the Tauride belt around the Kirsehir block (Fig. 1).

Paleomagnetic work conducted previously along the Aegean Arc, including on Lycian Taurides situated west of Isparta, has shown that its present-day curvature results from two successive rotational phases [1]. First, during the Early–Middle Miocene, rotations of about 20–25° in opposite directions affected the two terminations of the arc, initiating the curvature. Then, the structures of the western segment of the arc were brought to their present orientation by an additional Plio-Quaternary clockwise rotation of 25° [2]. These regional rotations imply a large amount of horizontal displacement, particularly in the central part of the arc. A recent evaluation based on new paleomagnetic data from Albania infers at least 250 km of westward displacement in the Gulf of Corinth [3] due to the total amount of about 45° clockwise rotation undergone by the western termination of the arc since Early Miocene times. In this case, outward displacement must be balanced by the extension observed in the core of the arc.

However, this tectonic model linking the back-arc extension to the rotation observed at the periphery of the Aegean Arc cannot be applied directly to the Tauride arc because no major coeval extension has so far been observed in its core. Consequently, it was important to extend paleomagnetic work in the Taurides in order to

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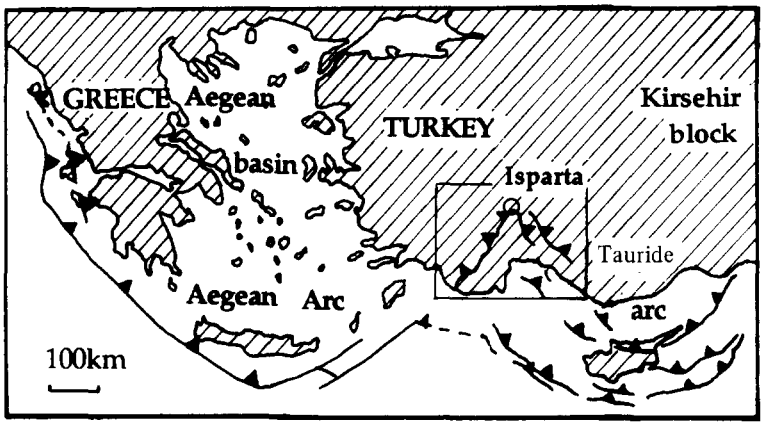


Fig. 1. Sketch map of the Aegean and Tauride arcs showing the location of the studied area.

provide a geodynamical framework in which the original configuration could be investigated and to advance our knowledge of arcuate structures generally.

This paper reports on the first paleomagnetic study carried out in the Cenozoic formations of the Western Taurides belt, in the region of Akseki–Beysehir and in the Ermenek basin (Figs. 2 and 3).

2. Geological setting and sampling

The studied area is located in the westernmost part of the Taurides orocline, east of the Antalya

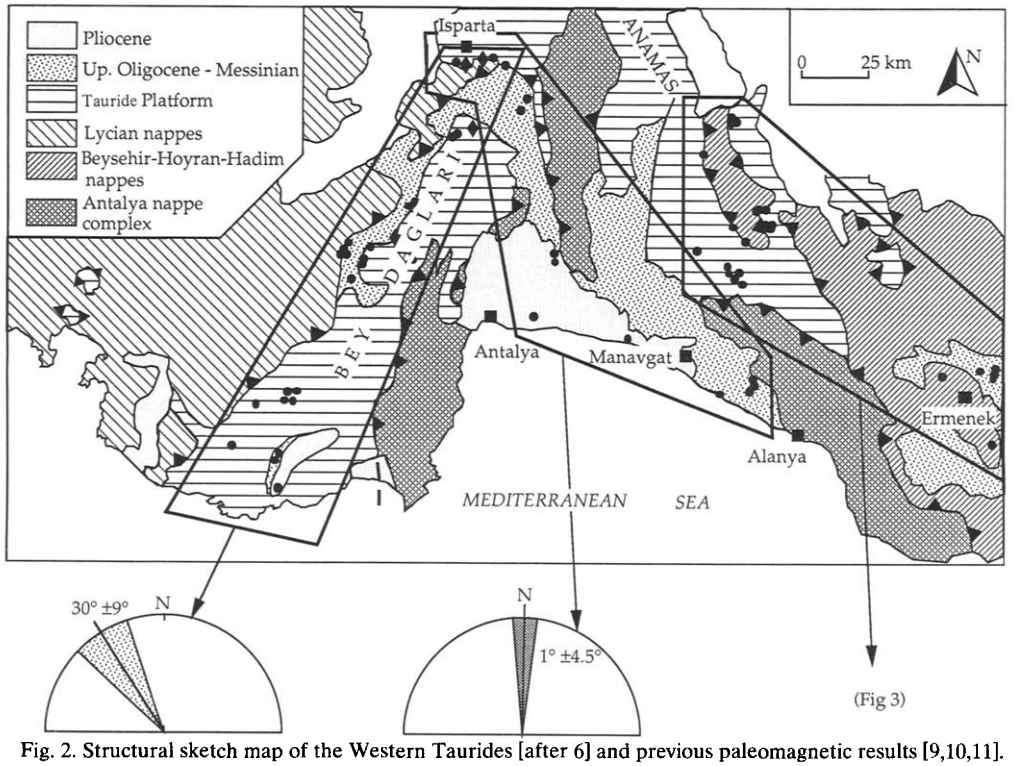


Fig. 2. Structural sketch map of the Western Taurides [after 6] and previous paleomagnetic results [9,10,11].

basin (Fig. 2). The characteristic large-scale tectonic feature of the region is a major reentrant known as the Isparta angle [4] which joins the Lycian Taurides, the eastward extension of the Aegean arc, and the Akseki–Beysehir Taurides (Western Taurides *sensu stricto*). The latter form part of the Tauride arc. Because of its key position within the Alpine Mediterranean belt and its complex structure, many investigations have been undertaken in this zone [5–8]. Generally, the Isparta angle is interpreted as the result of the interference between SE-vergent (Lycian Taurides) and SW-vergent (Akseki–Beysehir Taurides) allochthonous units emplaced during polyphase tectonic development from the Late Cretaceous to at least the Late Miocene [6]. The paleomagnetic results previously obtained in these regions are schematically shown in Fig. 2. Upper Cretaceous to Langhian sites distributed from the coast to Isparta consistently show that the Bey Daglari structure, located at the front of the Lycian nappes, has undergone a 30° post-Langhian counterclockwise rotation [9,10]. This rotation has been attributed by the authors to the Middle Miocene (Lycian) tectonic phase. In contrast, no significant rotation has affected the Antalya basin since at least the Langhian [11].

The Akseki–Beysehir Taurides comprise two major structural units (Fig. 3). The Beysehir–Hoyran–Hadim (BHH) nappes form a structurally complex association of Paleozoic and Mesozoic sedimentary units and ophiolitic rocks. These nappes regionally overlie a thick carbonate platform sequence known as the 'Tauride platform' or the 'Taurides autochthon'. This domain forms an arcuate thin-skinned fold-and-thrust belt in which the 'Tauride platform' carbonates and the overlying BHH nappes are folded together (Fig. 4).

The typical 'Tauride platform' succession consists of a thick Mesozoic succession (Triassic detrital formations, and Jurassic and Cretaceous limestones) and a thin Cenozoic succession (Paleocene to lower Eocene nummulitic limestones and middle Eocene flysch) overlying a Lower Paleozoic substratum (Cambrian limestones and Ordovician shales) [12].

The BHH nappe complex comprises various units in which the transition zones from a carbonate platform to a pelagic basin have been recognized within the Mesozoic facies [13]. The ophiolitic bodies top the sequence. Paleogeographic considerations thus provide evidence for the existence of a Neo-Tethyan oceanic domain north of the Taurides during Jurassic–Cretaceous times [6].

The geometry of the thrust belt is characteristic of a progressive piggy-back sequence that has propagated towards the southwest. In agreement with this regional sequence, stratigraphic data show that the obduction of the ophiolites occurred during the Late Cretaceous, whereas middle Eocene flysch is involved in the underlying thrust belt.

The tectonic style of the belt is illustrated by a general cross section (Fig. 4) based on a reinterpretation of the existing geological data [13]. We favour an interpretation in which the major structures are considered as ramp-related folds resulting from the existence of three major décollement levels. The deepest is situated at the base of the Cambrian carbonates and probably represents a general structural feature over the whole Tauride area. The second is located within the Triassic shales (Kasimlar Formation) and the highest emerges within the Eocene flysch, at the top of the sedimentary pile of the 'Tauride platform'.

In the same way, in the Seydisehir area, we interpret the outcropping of Triassic and Paleozoic formations as a large anticlinal culmination formed by two horses, refolding the roof strata and the basal contact of the nappes into a wide syncline (Figs. 3 and 4). This passive roof duplex geometry [14] is probably responsible for the back-thrusting of the cover strata at the leading edge of the blind culminations (Fig. 4).

At the southern border of the belt, the SW-vergent structures affecting the 'Tauride platform' disappear beneath the folded basal contact of the Antalya nappes and the Miocene deposits of the Antalya basin. The Antalya basin has been classically considered as a 'post-nappe' basin overlying the previously deformed terranes of the Taurides thrust belt of Akseki–Beysehir. However, the fact that the Antalya basin has been passively uplifted to form a SW-dipping monocline covering the front of the Taurides belt suggests the existence of a deep-seated intercutaneous wedge beneath the basin, probably beneath

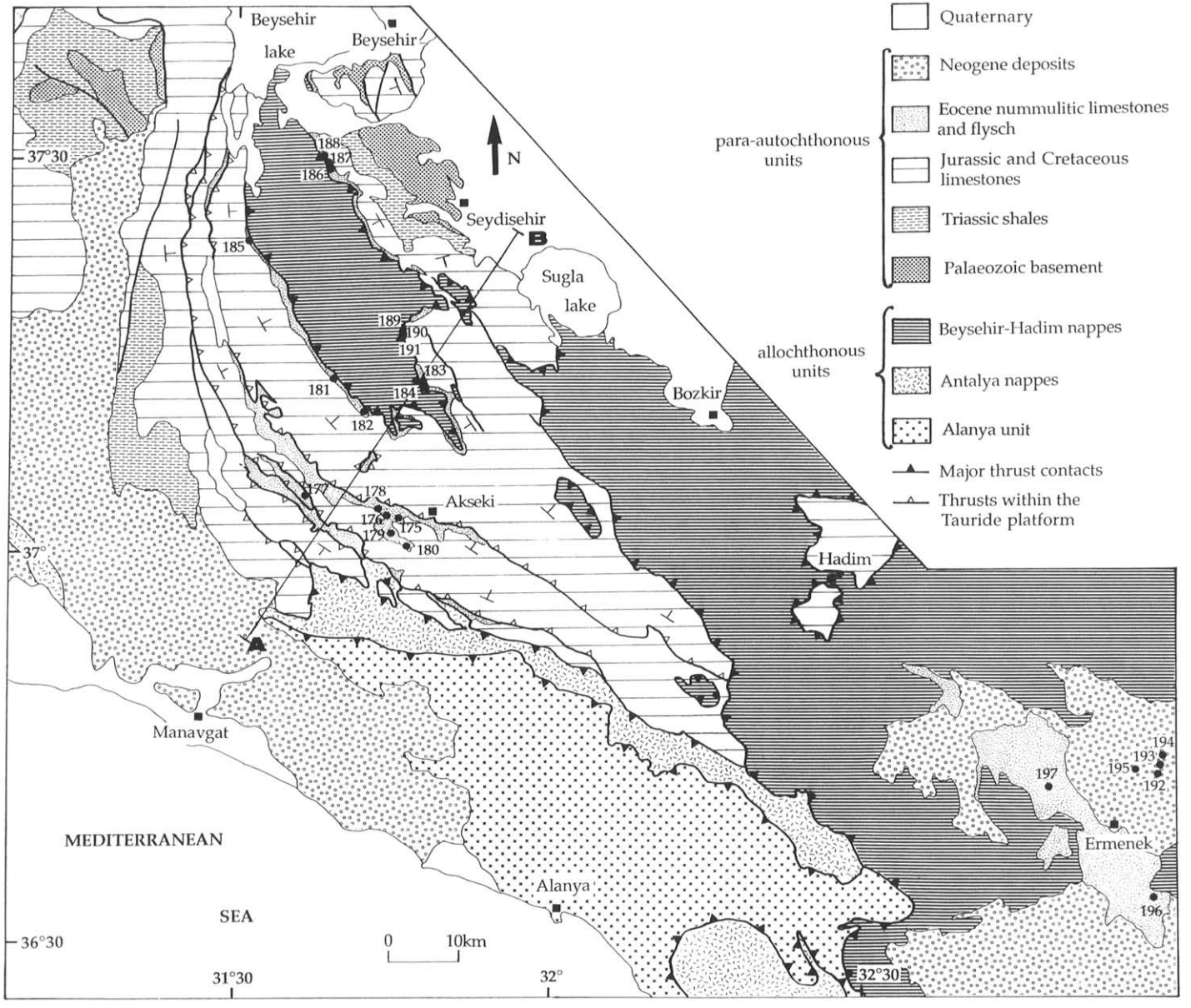


Fig 3. Geological map of the Taurides of Akseki–Beysehir up to the Ermenek basin with the location of the sampled sites [after 33,18]. Section line AB refers to the cross section of Fig. 4.

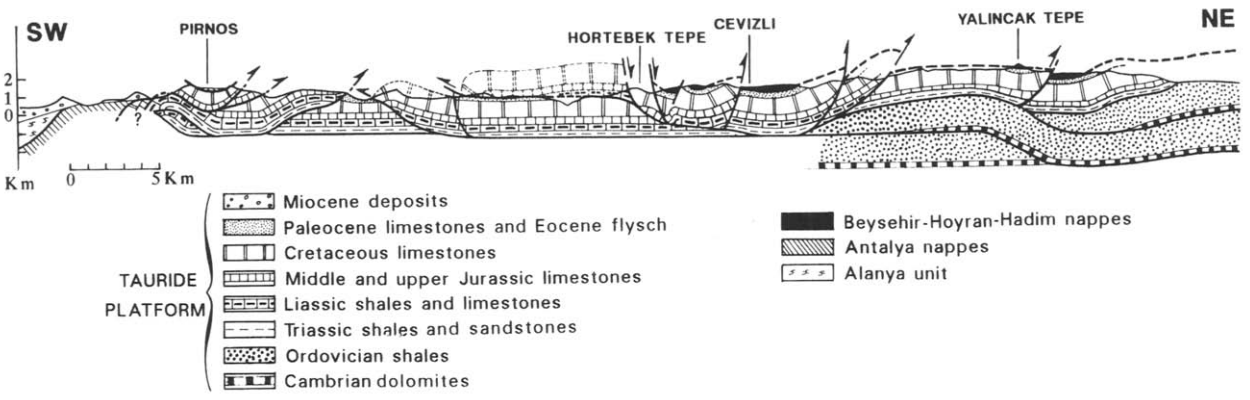


Fig. 4. NE–SW cross section through the Akseki thrust belt (section line in Fig. 3).

the basal contact of the Antalya nappes. NE-verging structures in the 'Tauride platform' may form as a consequence (Fig. 4). In this case, the building of the Taurides east of Antalya can no longer be considered as having been completely achieved before the Miocene. Moreover, the propagation of a major blind thrust fault beneath the basin during the Late Miocene is attested to by the existence of the Aksu thrust zone affecting the Tortonian deposits in the middle of the basin

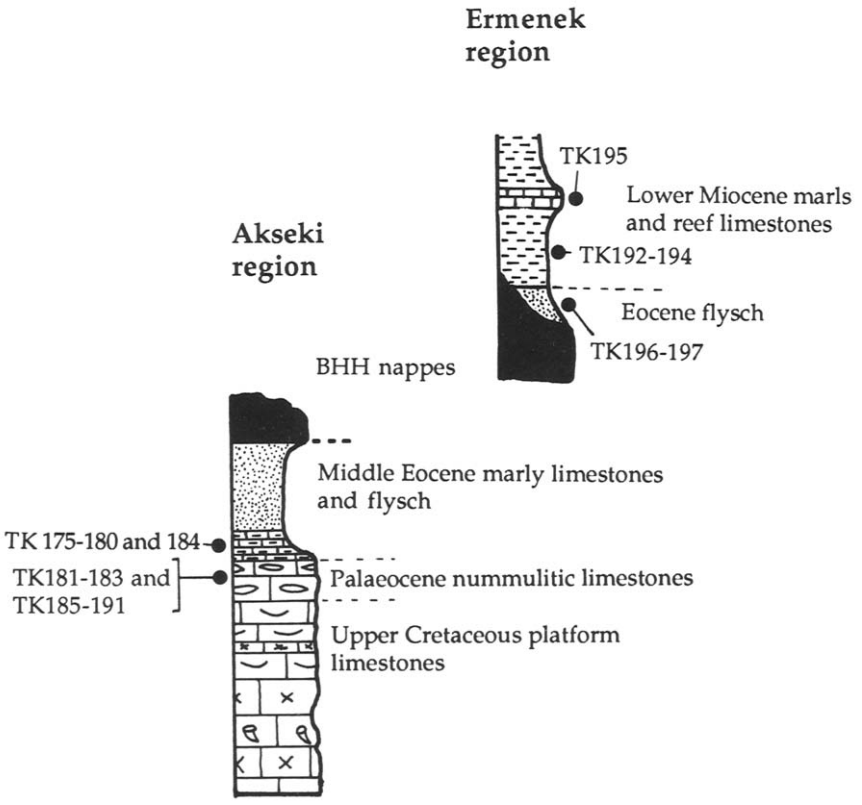


Fig. 5. Simplified stratigraphic column of the Cenozoic formations of the Akseki thrust belt [after 13] and location of the sampled sites.

[15]. The paleomagnetic results obtained previously indicate, however, that the movement along this thrust has not been accompanied in the Antalya basin by any significant differential rotation [11].

On the other hand, offshore seismic profiles have shown the existence of a S-verging thrust front in the Florence ridge, south of Antalya, in which the Messinian deposits are involved [16]. This implies that the Antalya basin as well as the whole Tauride structure was transported as a rigid body. The precise kinematics of this late thrusting are not known. However, shear criteria measured at the base of the Antalya nappes show a consistent post-Eocene south-southwesterly displacement [17].

Two major zones have been sampled for the paleomagnetic analysis. The precise location of the sampled sites is reported in Fig. 3.

The westernmost sampled area is the Beysehir and Akseki region. A total of seventeen sites were sampled in the Paleocene–Eocene formations of the 'Tauride platform' (Fig. 5). Sites TK175 to TK180 and TK184 were measured in pink to yellow marly limestones at the base of the middle Eocene flysch of Akseki. The other sites (TK181 to TK183 and TK185 to TK191) were sampled in Paleocene nummulitic limestones outcropping in the two limbs of the wide syncline supporting the Beysehir nappes.

The second sampling area corresponds to the Neogene Ermenek basin, situated at the eastern margin of the belt. This is an intramontane basin lying on the deformed terranes of the Hadim nappe (Fig. 3). It comprises undeformed marine deposits of Early–Middle Miocene age. Its structural position is very similar to that of the Antalya basin and consequently it was probably passively transported. Sites TK192 to TK194 have been sampled in grey-beige marly deposits of Lower Miocene age and TK195 was sampled in white reef limestones (Fig. 5). The marly layers in which TK196 and TK197 were sampled are very poor in microfauna. Following Blumenthal [18] we have attributed a pre-Miocene, (probably Eocene) age to these sites, in agreement with their basal stratigraphic position in the basin (Fig. 5).

Each site is located on an outcrop of 20–50 m in length in which a minimum of ten cores were taken. The cores are distributed all over the outcrop, both horizontally and vertically, in order to better average out the secular variation and to avoid possible errors due to very local effects. The localities with gentle bedding and consistent with regional structure were preferentially sampled. The cores were oriented *in-situ* using both a magnetic and a sun compass.

3. Laboratory measurements

3.1. Magnetic characteristics

The natural remanent magnetization (NRM) of the samples is generally low, ranging between about 20×10^{-6} A m^{-1} for the less magnetized Paleocene nummulitic limestones and Miocene Ermenek reef limestones and 8×10^{-3} A m^{-1} for the more strongly magnetized samples of the Akseki flysch.

In order to determine the nature of the minerals carrying the remanent magnetization, we investigated the stepwise acquisition of isothermal remanent magnetization (IRM) up to 1.5 T and the measurement of the remanent coercive field (H_{cr}) for one sample from each site. Typical IRM acquisition curves obtained from the different formations are shown in Fig. 6.

For most of the samples from the Akseki region (nummulitic limestones and marly limestones of the Akseki flysch), at least 80% of the saturation of the IRM is achieved in fields lower than 300 mT and H_{cr} does not exceed 40 mT, indicating that most of the magnetization is carried by a low-coercivity mineral such as mag-

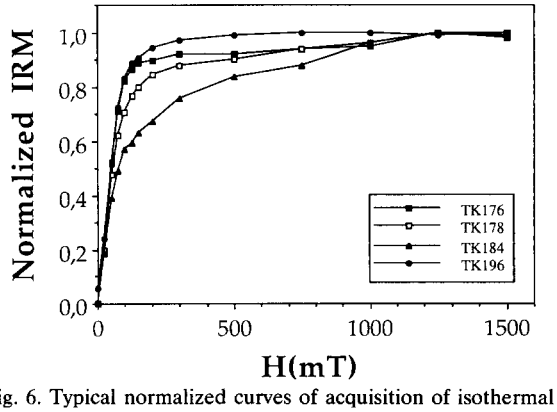


Fig. 6. Typical normalized curves of acquisition of isothermal remanent magnetization.

netite. This is confirmed by the magnetic behaviour of the samples from the same formations during thermal demagnetization characterized by blocking temperatures of about 550°C. However, for a few sites sampled in the pinker layers of marly limestones of the Akseki flysch (sites TK178 and TK184 in Fig. 6), the increase of IRM up to 1.5 T and values of H_{cr} up to 60 mT suggest the presence of a higher coercivity mineral, probably haematite, in addition to magnetite. In order to better investigate the physical properties of the magnetic mineralogy, we performed a hysteresis loop analysis on representative specimens of each formation using a Micromag 2900 Alternating Gradient Force Magnetometer. A curve is shown in Fig. 7 for a specimen of pink marly limestones from the Akseki flysch (TK178). The reversibility of the curve at 0.3 T indicates that haematite does not contribute significantly to the natural remanent magnetization. At higher fields, the slope of the curve, which at this point is due only to the paramagnetic contribution, is likely to be related to clay minerals. Moreover, the ratio of the hysteresis parameters corrected for the paramagnetic contribution (saturation of the ferromagnetic moment M_{fs}, saturation remanence M_{rs}, coercive field H_c and remanent coercive field H_{cr}) show that the remanent magnetization is

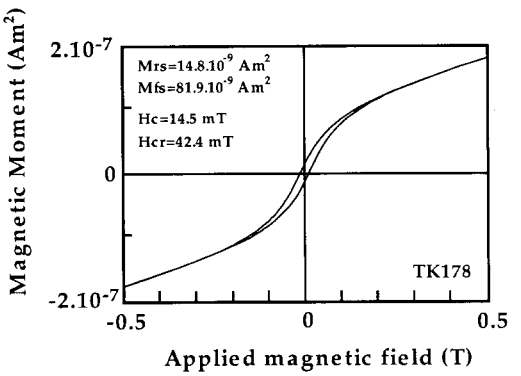


Fig. 7. Hysteresis curve obtained on a sample from the Eocene pink marly limestones of site TK178.

carried preferentially by pseudo-single-domain magnetite [19].

The samples from the Ermenek region present rather similar characteristics (saturation of the IRM at 300 mT (Fig. 6) and coercivity around 40 mT) but the blocking temperatures do not exceed 400°C. Moreover, during heating, the low-field susceptibility increases strongly after 350°C. Such behaviour is probably due to the presence of iron-sulphide minerals.

Anisotropy of magnetic susceptibility (AMS) was also measured, using a Kappabridge susceptibility bridge KLY2. The AMS ellipsoid reflects

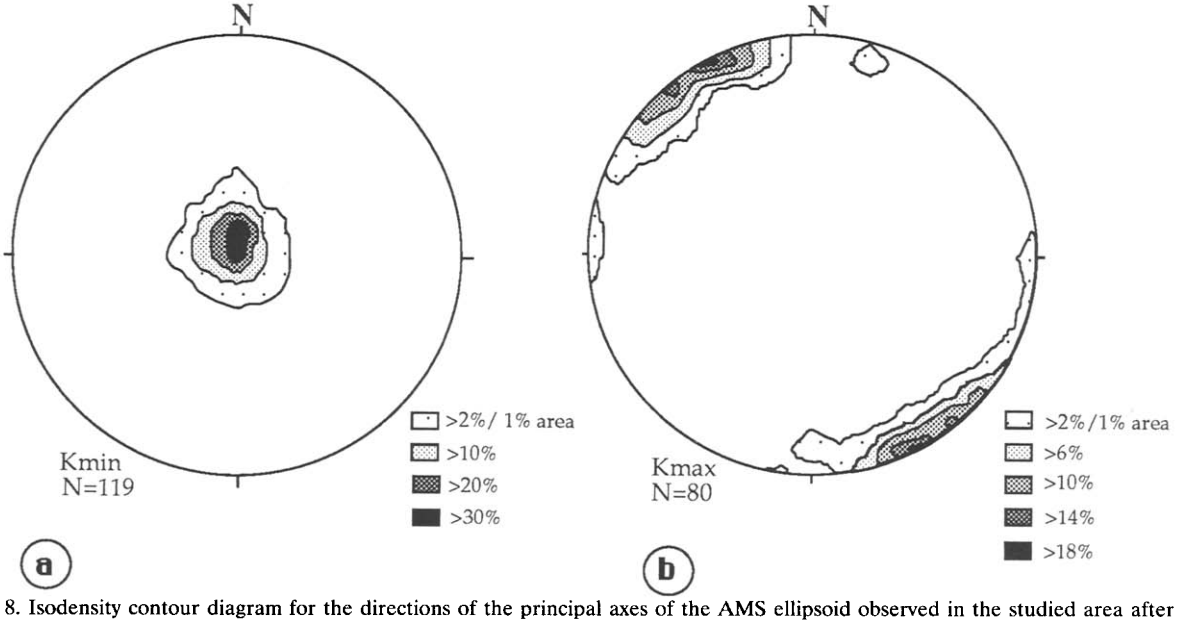


Fig. 8. Isodensity contour diagram for the directions of the principal axes of the AMS ellipsoid observed in the studied area after bedding correction. (a) Minimum axis K_{min}. (b) Maximum axis K_{max} at sites where a magnetic lineation is well defined.

the statistically preferred orientation of grains and/or crystal lattices of all the minerals contributing to the magnetic susceptibility. Consequently, AMS may provide helpful information on the depositional and deformational history of the sediments.

Except for the nummulitic and reef limestones samples, which show diamagnetic susceptibilities, the measurements revealed clearly anisotropic fabrics characterized by an oblate ellipsoid with the minimum axis K_{min} perpendicular to the bedding (Fig. 8a). This fabric, which is of typically

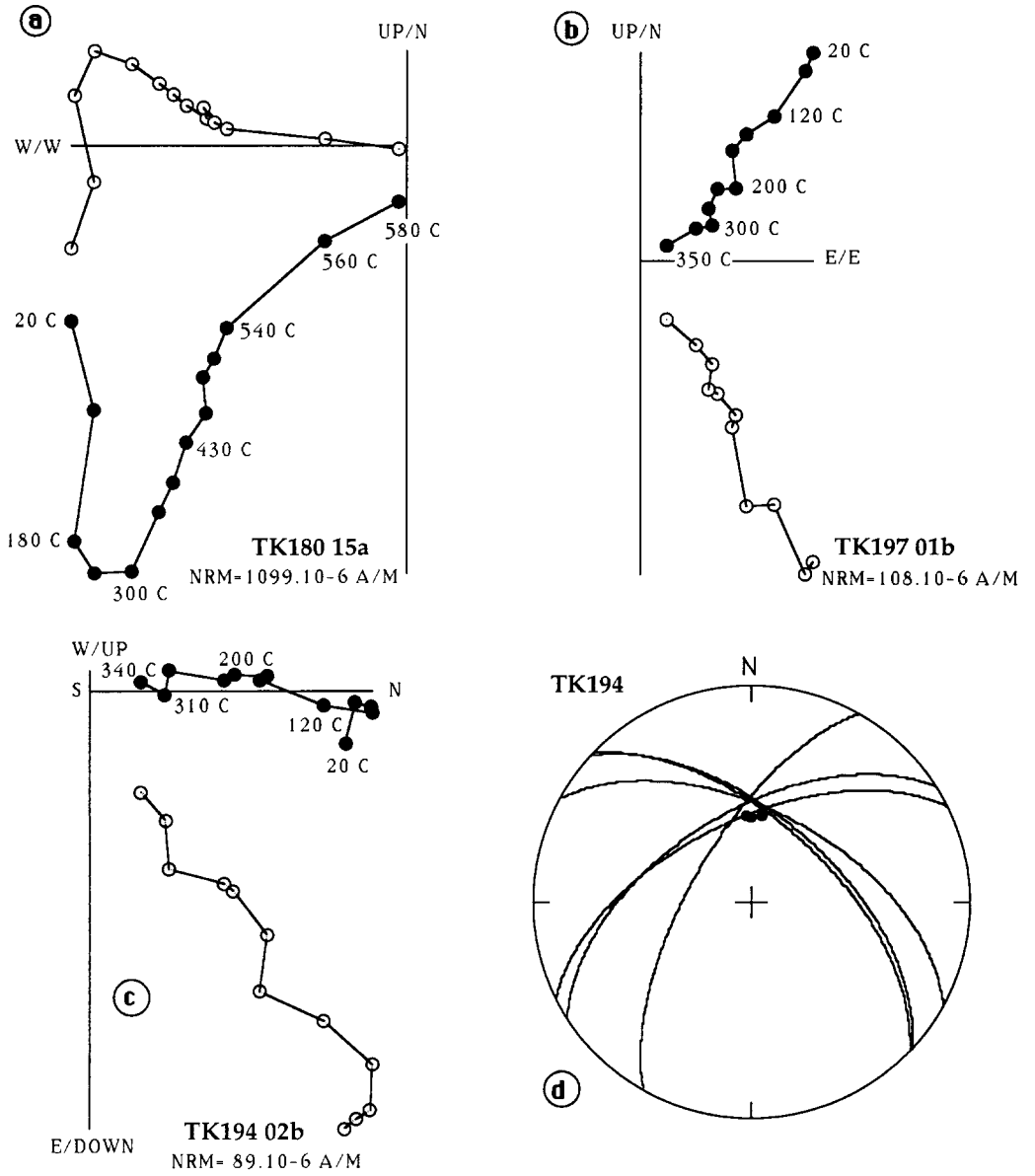


Fig. 9. Representative thermal demagnetization diagrams [34] of specimens from (a) Eocene pink marly limestones from the Akseki region, (b) Eocene marls from the Ermenek region and (c) Miocene marls from the Ermenek basin. All diagrams are corrected for bedding tilt if present. • = horizontal projection; ○ = vertical projection. (d) Equal-area lower hemisphere projection of remagnetization circles and stable directions obtained for a Miocene site with horizontal bedding from the Ermenek basin.

sedimentary origin, provides good evidence that the deposition of the studied formations occurred in a low-energy environment with no major disturbance. On the other hand, for all the sites sampled in the Eocene marly limestones, a clustering of the maximum axis K_{max} is observed within the bedding plane along the NW–SE direction, parallel to the regional trend of the structures (Fig. 8b). Such a feature has frequently been described in fold–thrust belts and generaly attributed to an early compressional stage aligning the K_{max} axes perpendicular to the shortening direction [20–23]. Thus, in our case, it may be interpreted as resulting from a regional NE–SW shortening. We observe that the directions of the magnetic lineation are better grouped than the local trends of the bedding. This suggests that this magnetic lineation is acquired before the folding occurred, and resulted from an early parallel layer shortening as already described in other fold–thrust structures [24]. In any case, degrees of anisotropy for all these sites are less than 10% and consequently cannot significantly bias the paleomagnetic directions.

3.2. Paleomagnetic data analysis

Thermal demagnetization was systematically carried out on one specimen per core with ten to eighteen steps from room temperature up to the limit of reproducible results (maximum 650°C). At each step of the thermal demagnetization, the low-field susceptibility was measured to monitor changes in the magnetic mineralogy. Measurements of the remanent magnetization were made in a shielded room using a 2G three-axis cryogenic magnetometer.

The general magnetic behaviour of the Eocene marly limestones from the Akseki–Beysehir region is illustrated in Fig. 9a. Three components are generally observed. The low-temperature component, which is removed below 300 °C, corresponds to the present-day dipolar direction of magnetization and is characteristic of a viscous component of magnetization. From 300°C up to about 550°C a component of reverse polarity is identified and can be accurately defined (Fig. 9a). As shown by the blocking temperatures, this latter is carried by magnetite and corresponds to the more stable component of magnetization. Within each site, the directions of these components have been averaged using Fisherian statistics [25] in order to obtain the local mean directions of magnetization. After 550°C, a high-temperature component carried by haematite was sometimes evident. However, its low intensity combined with other mineralogical changes revealed by an important increase in the low-field susceptiblity around 600°C does not allow us to clearly define its direction. A few samples at sites TK177 and TK178 have slightly different magnetic behaviour characterized by two subantipodal components, the thermal spectra of which largely overlap. For these samples, demagnetization diagrams fit on great circles, the intersection of which represents the common direction of magnetization [26,27]. The mean direction for these sites was calculated using the method described by McFadden and McElhinny [28] and generally corresponds to the stable component observed at the other sites from the same formation. The demagnetization paths of samples from site TK184, shown to display haematite in addition to magnetite (Fig. 6), fit on the same family of great circles so that no clear intersection can be defined. This behaviour is significant for two superimposed components of magnetization whose directions are non antipodal. However, it is worth noting that the great circles contain the sub-present-day direction of magnetization and the stable direction observed from sites of the same formation.

As far as the nummulitic limestones are concerned, most of the samples were too weakly magnetized, with NRMs of less than 30×10^{-6} A m^{-1}, to allow the definition of a stable component of magnetization. The direction of magnetization could be determined for only two sites (TK185 and TK190). In both cases, superimposed antipodal components were observed and remagnetization circles could be defined. For a few specimens of site TK190, a stable component of magnetization could be isolated and a mean local direction could be calculated together with the remagnetization circles using the method of McFadden and McElhinny [28].

Thermal demagnetization up to 300–400°C has been performed for the specimens sampled in the Ermenek basin, with the except ion of site TK195 sampled in reef limestones which was too weakly magnetized. For pilot samples, alternating field

demagnetization was also investigated and yielded similar results. For the Eocene sites (TK196 and TK197), a single stable component of magnetization was isolated up to 350–400°C (Fig. 9b). The transformation of the iron-sulphide minerals at higher temperatures did not permit us to continue the demagnetization process. Although the natural remanent magnetization was not always completely removed, we checked that the intensity decreased linearly to the origin so that the direction of the stable component of magnetization could always be accurately defined. For the other sites (TK192, TK193 and TK194), the demagnetization paths generally fit on great circles and sometimes reach a stable direction of magnetization (Fig. 9c and 9d).

4. Results

The mean declinations and inclinations calculated before and after bedding correction for each of the thirteen sufficiently well magnetized sites are reported in Table 1 and Fig. 10 with their respective statistical parameters.

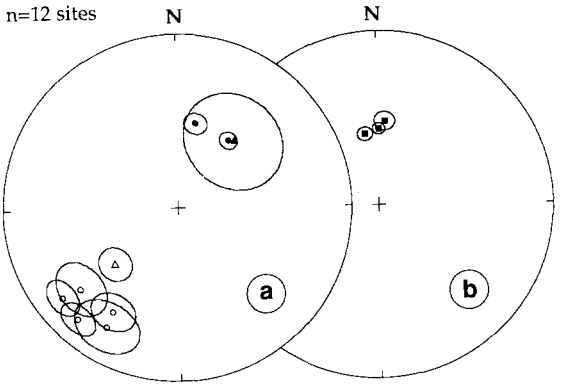


Fig. 10. Equal-area projection of the mean directions of magnetization with their relative 95% confidence circles obtained from each successful site from (a) Palaeocene and Eocene formations and (b) Miocene formations. Site TK175 has not been included because it has probably undergone a very local rotation. Black symbols = downward directions; white symbols = upward directions; △ and ▲ = Palaeocene sites; ○ and ● = Eocene sites; ■ = Miocene sites.

The eight Eocene sites gave the most satisfactory results (Table 1). The mean direction of magnetization at site TK175 significantly deviates

TABLE 1

Paleomagnetic directions reported as declination (D) and inclination (I) at each sufficiently magnetized site *in situ* ('is') and after bedding correction ('bc') and respective statistical parameters (K and α_{95}). The dip and dip direction of the bedding are also shown

Site	Age	N	D_{is}	I_{is}	D_{bc}	I_{bc}	K	α_{95}	Dip	Dip Dir
Akseki thrust belt										
TK175 *	Eoc.	10	4	38	4	17	34	7.5	21	005
TK176	Eoc.	9	232	−36	223	−15	38	7.5	28	005
TK177	Eoc.	7(7)	232	1	231	−28	27	11.9	30	245
TK178	Eoc.	8(1)	226	−36	212	−21	18	13.6	30	338
TK179	Eoc.	7	234	−11	233	−17	47	7.7	10	260
TK180	Eoc.	10	211	−45	213	−30	21	9.6	15	042
TK185	Paleoc.	9(9)	229	−50	233	−6	45	7.8	50	060
TK190	Paleoc.	5(3)	41	48	–	–	14	23	–	–
Ermenek basin										
TK192	Mioc.	11(11)	5	50	–	–	97	4.6	–	–
TK193	Mioc.	8(8)	350	56	–	–	268	3.4	–	–
TK194	Mioc.	9(6)	1	54	–	–	369	2.8	–	–
TK196	Eoc.	11	20	37	13	49	79	5.2	14	230
TK197	Eoc.	12	4	59	38	50	114	4.1	25	090
Mean paleomagnetic direction:										
Eoc.		$n = 7$	$D = 39$		$I = 30$		$K = 20.1$		$\alpha_{95} = 13.8$	
Eoc. + Paleoc.		$n = 9$	$D = 40$		$I = 35$		$K = 20.7$		$\alpha_{95} = 11.6$	
Mioc.		$n = 3$	$D = 359$		$I = 53$		$K = 216.0$		$\alpha_{95} = 8.4$	

N = number of samples used for statistics; ( ) = number of remagnetization circles among the total number of specimens used for statistics; * site rejected from the mean direction calculation

from the mean clustering of the other Eocene directions of magnetization. This may be due to a very local rotation and this site is rejected by the statistical calculation at the 2σ level. A statistical test of the scatter of the other data before and after bedding correction was performed using the method of McElhinny [29]. The test is positive at the 99% significance level, indicating that the magnetization was acquired before folding ($N = 7$, $K1/K2 = 6.87$, $F(12,12)1\% = 4.16$). We observe that the normal directions of magnetization at the two Eocene sites from the Ermenek basin show steeper inclinations than those of the reversed middle Eocene directions of magnetization (Fig. 10a and Table 1). This difference cannot be attributed to unsatisfactory cleaning of a normal polarity overprint because some of the mean magnetizations have been determined using both stable directions and remagnetization circles. We have noted above that the Eocene age of the sites from the Ermenek basin is not precisely assessed. Because these regions underwent a general northward drift during the early Tertiary, the steeper inclinations may be linked to a younger age of the Ermenek sites. However, whatever their age all these sites show similar values of declination. The mean regional direction calculated from these sites is: $N = 7$, $D = 39°$, $I = 30°$, $\alpha_{95} = 13.8°$.

Only two of the eleven sites measured in Paleocene limestones in the two limbs of the Beysehir syncline gave suitable results. One of the sites (TK190) is characterized by a horizontal bedding attitude but its mean direction of magnetization is exactly antiparallel to that of site TK185 before bedding correction. These two sites thus fail the fold test, suggesting that they have been remagnetized after the fold had been formed. The formation of authigenic magnetite favoured by fluid circulation during the deformation as suggested in the same region by Morris and Robertson [30] may be at the origin of this remagnetization, the age of which is not constrained.

In any case, the mean directions of magnetization of the two Paleocene sites before bedding correction show eastward deviated declinations of about 40°, similar to the mean declination calculated from the Eocene sites. This indicates that Paleocene and Eocene sites recorded the same rotational event that occurred after the late Eocene–Oligocene folding phase (or possibly during the deformational event).

A regional mean paleomagnetic direction has thus been calculated, averaging out the Paleocene directions in the geographic frame and the seven Eocene directions after bedding correction using Fisherian statistics [25]: $N = 9$, $D = 40°$, $I = 35°$, $\alpha_{95} = 11.5°$.

This direction illustrates a clockwise rotation of this area of about 40°, probably since late Eocene–Oligocene times.

The Lower Miocene sites of the Ermenek basin yield a mean regional paleomagnetic direction close to that of the present day: $N = 3$, $D = 359°$, $I = 53°$, $\alpha_{95} = 8.41°$. Because the bedding of these sites is horizontal, no fold test could be applied and we cannot exclude a recent remagnetization. However, the absence of rotation of this basin since the early Miocene is sustained by similar results obtained previously in the Antalya basin for the same period and where both positive fold and reversal tests attest to a primary magnetization [11].

5. Discussion

The paleomagnetic results presented above show that the Taurides of Akseki–Beysehir have undergone a post-middle Eocene clockwise rotation of about 40°. As in the Antalya basin (Fig. 1) [11], the Lower Miocene deposits of the Ermenek basin seem not to have been affected by any significant rotational movement. Thus the rotation measured in the Akseki formation occurred prior to the Early Miocene, during late Eocene or Oligocene time.

Although our data are not well enough distributed geographically to record variations in rotations along the regional strike of the belt we do note that no significant differential rotation is observed between sites sampled in different thrust sheets. We believe that this is a regional rotation affecting the entire Akseki–Beysehir thrust belt as far as the Ermenek region.

Considering a 40° rigid rotation of the belt about a 'pivot zone' situated few kilometres north of Isparta, the horizontal displacement may be evaluated at 103 km ± 26 km at the latitude of Akseki. The minimum shortening measured along the NW–SE cross section through the Akseki

thrust belt shown in Fig. 4 is about 20 km, which is much less than the 'rigid' displacement calculated above. These calculations are purely indicative, but they do suggest that the major part of the required transport should be accommodated by a deeper seated décollement fault dying out far away into the Mediterranean Sea. This is in agreement with the fact that the 35–40° clockwise rotation occurred after the thrust-related folding developed in the Akseki–Beysehir area (TK185 and TK190).

These results complete the picture of the complex geodynamical evolution of the Western Taurides belt during the Alpine orogenesis.

As suggested by Kissel and Poisson [10], the difference between the Middle Miocene 30° anticlockwise rotation of the Bey Daglari and the insignificant rotation of the Antalya basin (Fig. 2) implies the existence of a major decoupling zone between the Lycian and Akseki–Beysehir Taurides, probably at the base of the Antalya nappe complex. Together with the previously published data [9,10,11], these new results lead us to propose a preliminary sketch for the formation of the Isparta reentrant (Fig. 11) that is very similar to that recently proposed for the Northwest Himalayan syntaxes [31]. Its present-day configuration would have resulted from diachronous rotational displacements, in opposite senses, of the Lycian Taurides and the Akseki–Beysehir Taurides:

(i) During the late Eocene–Oligocene, a 40° clockwise rotation of the Akseki–Beysehir Taurides developed.

(ii) During the Middle Miocene, the Lycian Taurides along with the Tauride platform formations situated west of the Antalya nappes (Western Bey Daglari) underwent a 30° anticlockwise rotation. This later motion is probably accommodated by underthrusting beneath the Antalya nappe complex and the Tauride platform formations situated east of Antalya (Eastern Bey Daglari) as suggested by Marcoux et al. [17].

These paleomagnetic results still need to be confirmed and extended regionally. In order to complete the tectonic outline of the Tauride arc, two major regions require new paleomagnetic and tectonic work: the Eastern Taurides forming the eastern 'limb' of the Tauride arc, and the Kirsehir block (Fig. 1) situated in the core of the

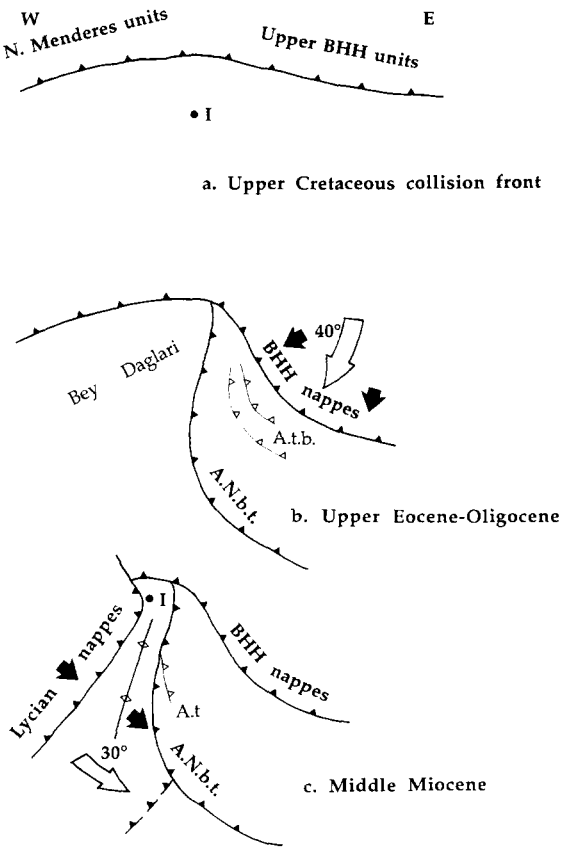


Fig. 11. Schematic model for the formation of the Isparta reentrant. See explanation in text. *BHH nappes* = Beysehir–Hoyran–Hadim nappes; *A.N.b.t.* = Antalya nappe basal thrust; *A.t.b.* = Akseki thrust belt; *A.t.* = Aksu thrust; *I* = Isparta.

arc. However, from now on the existence of important rotational movements of the Tauride belt implies that this latter 'crustal block' may no longer be considered as having been completely rigid during the Neogene, as has been pointed out by Alparslan et al. [32] on the basis of structural data.

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